

## 10.10 Orthogonal Polynomials

Section 2 ran Gram–Schmidt on  $(1, x, x^2)$  with  $\langle p, q \rangle = \int_{-1}^1 pq$ , and stopped at degree 2 only because the space did (Example (Gram–Schmidt on  $1, x, x^2$ )). Nothing in the process cared. Feed it  $(1, x, x^2, x^3, \dots)$  and it returns an infinite orthogonal sequence, one polynomial of each degree. Three things make such sequences worth a section. Each member is computable from the two before it. Each member of degree  $k$  has  $k$  distinct real roots inside the interval. And those roots are the best possible points at which to sample a function to estimate its integral. The first fact rests on one small identity; the other two rest on the first.

Throughout this section the field is  $\mathbb{R}$ .

### 10.10.1 Two ways to make an inner product on polynomials

Both constructions we need appeared in Section 1.

**From a weight.** Fix an interval with endpoints  $-\infty \leq a < b \leq \infty$  and a **weight**  $w$ : a function continuous and **strictly positive** on the open interval  $(a, b)$ , with  $\int_a^b |t|^k w(t) dt$  finite for every  $k \in \mathbb{N}$ . Define

$$\langle p, q \rangle = \int_a^b p(t) q(t) w(t) dt.$$

Finiteness of the moments makes this a real number for all  $p, q \in \mathbb{R}[x]$ , and bilinearity and symmetry come from linearity of the integral. For positivity, let  $p \neq 0$ . Then  $p^2 w$  is continuous and  $\geq 0$  on  $(a, b)$ , and  $> 0$  off the finitely many roots of  $p$  (Corollary 5.1.4), so  $\langle p, p \rangle > 0$ . That last step is the one calculus fact this section leans on: a continuous non-negative function on an interval whose integral is 0 vanishes identically. As in Definition 10.1.1 it follows from the local fact that a continuous function non-zero at a point stays bounded away from 0 on a small interval around it, which then contributes a positive amount to the integral.

**From nodes.** Fix distinct  $c_0 < \dots < c_n$  in  $\mathbb{R}$  and put  $\langle p, q \rangle = \sum_{i=0}^n p(c_i) q(c_i)$ . Section 1 checked that this is an inner product **on**  $\mathbb{R}[x]_{\leq n}$ , positivity coming from the root bound. It is not one on  $\mathbb{R}[x]$ : the polynomial  $(x - c_0) \cdots (x - c_n)$  is non-zero and pairs with itself to give 0. So in the node case everything below stops at degree  $n$ .

The two constructions share one property, and it is the engine of the section.

#### Proposition 10.10.1. Multiplication by $x$ is symmetric

Let  $\langle \cdot, \cdot \rangle$  be an inner product on  $\mathbb{R}[x]$  coming from a weight, or the node inner product on  $\mathbb{R}[x]_{\leq n}$ . Then

$$\langle xp, q \rangle = \langle p, xq \rangle$$

for all polynomials  $p, q$  in the space (in the node case, for all  $p, q$  of degree at most  $n - 1$ , so that  $xp$  and  $xq$  still lie in it).

*Proof.* In the weight case both sides equal  $\int_a^b t p(t) q(t) w(t) dt$ , and in the node case both sides equal  $\sum_{i=0}^n c_i p(c_i) q(c_i)$ . ■

The proof is one line, but the statement is not a triviality: multiplication by  $x$  moves from one slot to the other for free. That is exactly the identity  $\langle T\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, T\mathbf{v} \rangle$  defining a **self-adjoint** operator (Definition 10.6.2), and it is the reason the recurrence below has three terms instead of  $k + 2$ .

*Warning.* **Multiplication by  $x$  is not an operator on  $\mathbb{R}[x]_{\leq n}$ .** It raises degree, so it maps that space out of itself; on  $\mathbb{R}[x]$  it is an operator, but there the space is infinite-dimensional, and Definition 10.6.2 was stated for a finite-dimensional one. “Self-adjoint” is here a name for [Proposition 10.10.1](#), not a literal application of the definition. It is the identity, not the name, that the proofs use.

## 10.10.2 Orthogonal polynomial sequences

Applied to the independent list  $(1, x, x^2, \dots)$ , Gram–Schmidt never stalls. What comes out is one polynomial of each degree, pairwise orthogonal, and that combination deserves a name.

*An orthogonal polynomial sequence is one polynomial of every degree, no two of them pairing to anything but zero.*

### Definition 10.10.1. Orthogonal Polynomial Sequence

Let  $\langle \cdot, \cdot \rangle$  be an inner product on  $\mathbb{R}[x]$ , or on  $\mathbb{R}[x]_{\leq n}$ . An **orthogonal polynomial sequence** for it is a sequence  $p_0, p_1, p_2, \dots$  of polynomials (a finite list  $p_0, \dots, p_n$  in the second case) such that

(OP1)  $\deg p_k = k$  for **every** index  $k$ ;

(OP2)  $\langle p_j, p_k \rangle = 0$  whenever  $j \neq k$ .

The sequence is **monic** if every  $p_k$  has leading coefficient 1.

In words. (OP1) says one polynomial of each degree: no gaps, no repeats, no accidental drop in degree. (OP2) says the members are pairwise orthogonal. Together they say more than either alone, because by (OP1) the list  $(p_0, \dots, p_{k-1})$  is a basis of  $\mathbb{R}[x]_{\leq k-1}$ , so (OP2) upgrades to

$$\langle p_k, q \rangle = 0 \quad \text{for every } q \in \mathbb{R}[x]_{\leq k-1},$$

and that is the form in which the condition always gets used.

Two things need checking: that such a sequence exists, and that fixing the scale pins it down.

### Proposition 10.10.2. Existence and uniqueness of the monic sequence

Let  $\langle \cdot, \cdot \rangle$  be an inner product on  $\mathbb{R}[x]$ . For each  $k \in \mathbb{N}$  there is **exactly one** monic  $p_k$  of degree  $k$  with  $\langle p_k, q \rangle = 0$  for every  $q \in \mathbb{R}[x]_{\leq k-1}$ . These  $p_k$  form the unique monic orthogonal polynomial sequence, and  $(p_0, \dots, p_k)$  is an orthogonal basis of  $\mathbb{R}[x]_{\leq k}$  for every  $k$ . The same holds on  $\mathbb{R}[x]_{\leq n}$  for  $k \leq n$ .

*Idea.* Existence is Gram–Schmidt, once we know the  $k$ -th output really has degree  $k$ ; it does, because it lies in  $\mathbb{R}[x]_{\leq k}$  but not in the span of its predecessors, which is  $\mathbb{R}[x]_{\leq k-1}$ . Uniqueness is the standard move: two candidates differ by a polynomial of lower degree that

is orthogonal to everything of lower degree, hence to itself.

*Proof. Existence.* The list  $(1, x, \dots, x^k)$  is independent, so Theorem 10.2.2 applies and returns an orthonormal list  $(\mathbf{e}_0, \dots, \mathbf{e}_k)$  with  $\text{span}(\mathbf{e}_0, \dots, \mathbf{e}_j) = \text{span}(1, x, \dots, x^j) = \mathbb{R}[x]_{\leq j}$  for every  $j \leq k$ . Fix  $j$ . Then  $\mathbf{e}_j \in \mathbb{R}[x]_{\leq j}$ , and  $\mathbf{e}_j \notin \mathbb{R}[x]_{\leq j-1} = \text{span}(\mathbf{e}_0, \dots, \mathbf{e}_{j-1})$ , since an orthonormal list is independent (Theorem 10.1.4). Hence  $\deg \mathbf{e}_j = j$  exactly. Let  $a_j \neq 0$  be its leading coefficient and put  $p_j = \mathbf{e}_j/a_j$ , monic of degree  $j$ . For  $q \in \mathbb{R}[x]_{\leq j-1} = \text{span}(\mathbf{e}_0, \dots, \mathbf{e}_{j-1})$  we get  $\langle p_j, q \rangle = 0$  by orthogonality of the list.

*Uniqueness.* Suppose  $p$  and  $p'$  are both monic of degree  $k$  and orthogonal to every element of  $\mathbb{R}[x]_{\leq k-1}$ . Their leading terms cancel, so  $p - p' \in \mathbb{R}[x]_{\leq k-1}$ . Therefore  $\langle p - p', p - p' \rangle = \langle p, p - p' \rangle - \langle p', p - p' \rangle = 0$ , and positive definiteness gives  $p = p'$ .

Finally,  $(p_0, \dots, p_k)$  is an orthogonal list of non-zero vectors, hence independent (Theorem 10.1.4), with  $k + 1 = \dim \mathbb{R}[x]_{\leq k}$  members, hence a basis (Theorem 1.6.4). ■

**Examples.** For  $\langle p, q \rangle = \int_{-1}^1 pq$  the first three monic members are  $1, x, x^2 - \frac{1}{3}$  (Example (Gram–Schmidt on  $1, x, x^2$ )); these are the monic **Legendre polynomials**. For the node inner product with nodes  $-1, 0, 1$  on  $\mathbb{R}[x]_{\leq 2}$  we have  $\langle 1, 1 \rangle = 3$ ,  $\langle x, 1 \rangle = 0$ ,  $\langle x, x \rangle = 2$ ,  $\langle x^2, 1 \rangle = 2$ ,  $\langle x^2, x \rangle = 0$ , so  $p_0 = 1$ ,  $p_1 = x$ ,  $p_2 = x^2 - \frac{2}{3}$  — and there the sequence ends, since  $\mathbb{R}[x]_{\leq 2}$  holds nothing of degree 3. The degenerate member is always  $p_0 = 1$ , the only monic polynomial of degree 0, with an empty condition on it.

**A non-example by minimal change.** Replace the Legendre  $p_2 = x^2 - \frac{1}{3}$  by  $\tilde{p}_2 = x^2 + x - \frac{1}{3}$ . Still monic of degree 2, so (OP1) survives; but  $\langle \tilde{p}_2, x \rangle = \langle x, x \rangle = \frac{2}{3} \neq 0$ , so (OP2) fails. Adding a multiple of a lower member is exactly the freedom orthogonality removes.

**Why monic?** Rescaling each  $p_k$  by a non-zero  $\lambda_k$  gives another orthogonal polynomial sequence, so a normalization has to be chosen, and traditions differ: the classical Legendre polynomials  $P_k$  are fixed by  $P_k(1) = 1$ , and Gram–Schmidt as run in Section 2 fixes  $\|p_k\| = 1$ . Monic is the convenient choice here, because it is the one for which the recurrence below needs no extra factor.

*Warning.* **The sequence belongs to the inner product, not to the polynomials.** “The orthogonal polynomials” is meaningless without a weight and an interval. Exercise B2 of Section 2 (Exercise (B2: Gram–Schmidt on polynomials)) ran the same process with  $\int_0^1$  instead of  $\int_{-1}^1$  and got  $1, x - \frac{1}{2}, x^2 - x + \frac{1}{6}$ : same degrees, different polynomials.

*Quick check.* For  $\langle p, q \rangle = \int_{-1}^1 pq$ , how many polynomials of degree 2 with  $\|p\| = 1$  are orthogonal to  $\mathbb{R}[x]_{\leq 1}$ ? Write them down.

*Solution.* Exactly two. If  $p$  has degree 2 and  $p \perp \mathbb{R}[x]_{\leq 1}$ , let  $\lambda \neq 0$  be its leading coefficient; then  $p/\lambda$  is monic of degree 2 and still orthogonal to  $\mathbb{R}[x]_{\leq 1}$ , so  $p/\lambda = x^2 - \frac{1}{3}$  by [Proposition 10.10.2](#). Thus  $p = \lambda(x^2 - \frac{1}{3})$ , and  $\|p\| = 1$  leaves  $\lambda = \pm 1/\|x^2 - \frac{1}{3}\|$ . Since Example

(Gram–Schmidt on  $1, x, x^2$ ) computed  $\|x^2 - \frac{1}{3}\|^2 = \frac{8}{45}$ , the two answers are  $\pm \frac{\sqrt{10}}{4}(3x^2 - 1)$ . Fixing the scale by the norm leaves a sign; fixing it by “monic” leaves nothing.

### 10.10.3 The three-term recurrence

Computing  $p_n$  by Gram–Schmidt costs one inner product for every earlier member, so reaching degree  $n$  costs about  $n^2/2$  integrals. Almost all of them return 0. The next theorem says which ones do not, and reduces the cost to two integrals per step.

#### Theorem 10.10.3. Three-Term Recurrence

Let  $(p_k)$  be the monic orthogonal polynomial sequence of [Proposition 10.10.2](#), for an inner product satisfying [Proposition 10.10.1](#). For a weight on an interval this covers every  $k$ ; for the node inner product on  $n$  nodes, where [Proposition 10.10.1](#) holds only up to degree  $n-1$ , read the statement for  $k+1 \leq n$ . Put  $p_{-1} = 0$  and, for  $k \geq 0$ ,

$$a_k = \frac{\langle xp_k, p_k \rangle}{\langle p_k, p_k \rangle}, \quad b_k = \frac{\langle p_k, p_k \rangle}{\langle p_{k-1}, p_{k-1} \rangle} \quad (k \geq 1).$$

Then  $p_0 = 1$ ,  $p_1 = x - a_0$ , and

$$p_{k+1} = (x - a_k)p_k - b_k p_{k-1} \quad \text{for every } k \geq 1.$$

Moreover  $b_k > 0$  for every  $k \geq 1$ .

*Idea.* The polynomial  $xp_k$  is monic of degree  $k+1$ , so it is  $p_{k+1}$  plus something of degree at most  $k$ , and that something expands in the orthogonal basis  $(p_0, \dots, p_k)$  with coefficients  $\langle xp_k, p_j \rangle / \langle p_j, p_j \rangle$  ([Corollary 10.3.4](#)). The point is that almost every coefficient is 0: move the  $x$  across with [Proposition 10.10.1](#), and the coefficient becomes  $\langle p_k, xp_j \rangle$ , which vanishes as soon as  $\deg(xp_j) = j+1$  is still below  $k$ . Only  $j = k$  and  $j = k-1$  can survive.

*Proof.*  $p_0 = 1$  is the unique monic polynomial of degree 0, and  $p_1 = x - a_0$  is the unique monic polynomial of degree 1 with  $\langle p_1, p_0 \rangle = 0$ , since  $\langle x - a_0, 1 \rangle = \langle x, 1 \rangle - a_0 \langle 1, 1 \rangle = 0$  exactly for  $a_0 = \langle x \cdot 1, 1 \rangle / \langle 1, 1 \rangle$ .

Now fix  $k \geq 1$ . By [Proposition 10.10.2](#),  $(p_0, \dots, p_{k+1})$  is an orthogonal basis of  $\mathbb{R}[x]_{\leq k+1}$ , and  $xp_k$  lies in that space. By [Corollary 10.3.4](#) applied inside  $\mathbb{R}[x]_{\leq k+1}$ ,

$$xp_k = \sum_{j=0}^{k+1} c_j p_j, \quad c_j = \frac{\langle xp_k, p_j \rangle}{\langle p_j, p_j \rangle}.$$

Both  $xp_k$  and  $p_{k+1}$  are monic of degree  $k+1$ , and every other term of the sum has degree at most  $k$ ; comparing coefficients of  $x^{k+1}$  gives  $c_{k+1} = 1$ .

Let  $j \leq k-2$ . By [Proposition 10.10.1](#),  $\langle xp_k, p_j \rangle = \langle p_k, xp_j \rangle$ , and  $\deg(xp_j) = j+1 \leq k-1$ , so  $xp_j \in \mathbb{R}[x]_{\leq k-1}$  and the pairing is 0 by [Proposition 10.10.2](#). Hence  $c_j = 0$  for every  $j \leq k-2$ , and

$$xp_k = p_{k+1} + c_k p_k + c_{k-1} p_{k-1},$$

which rearranges to  $p_{k+1} = (x - c_k)p_k - c_{k-1}p_{k-1}$ .

It remains to identify the two coefficients. By definition  $c_k = \langle xp_k, p_k \rangle / \langle p_k, p_k \rangle = a_k$ . For the other,  $xp_{k-1}$  is monic of degree  $k$ , so  $xp_{k-1} - p_k$  has degree at most  $k-1$  and therefore pairs to 0 with  $p_k$ . Hence, using [Proposition 10.10.1](#) again,

$$\langle xp_k, p_{k-1} \rangle = \langle p_k, xp_{k-1} \rangle = \langle p_k, p_k \rangle,$$

so  $c_{k-1} = \langle p_k, p_k \rangle / \langle p_{k-1}, p_{k-1} \rangle = b_k$ . Finally  $b_k$  is a quotient of  $\|p_k\|^2$  by  $\|p_{k-1}\|^2$ , both strictly positive because  $p_k$  and  $p_{k-1}$  are non-zero, so  $b_k > 0$ . ■

Two inner products per step generate the whole sequence. The positivity of  $b_k$  is not decoration either: it is the sign that drives the interlacing argument of [Exercise \(C1: Interlacing roots\)](#).

*Remark.* If the interval is symmetric,  $(a, b) = (-c, c)$ , and the weight is even,  $w(-t) = w(t)$ , then  $a_k = 0$  for every  $k$  and  $p_k(-x) = (-1)^k p_k(x)$ . Induct on  $k$ :  $p_0 = 1$  is even, and  $a_0 = \langle x, 1 \rangle / \langle 1, 1 \rangle = 0$  because  $tw(t)$  is odd, so  $p_1 = x$  is odd. If  $p_{k-1}$  and  $p_k$  have parities  $(-1)^{k-1}$  and  $(-1)^k$ , then  $tp_k(t)^2 w(t)$  is odd, so  $a_k = 0$ , and  $p_{k+1} = xp_k - b_k p_{k-1}$  is a combination of two polynomials of parity  $(-1)^{k+1}$ .

### Example. The monic Legendre polynomials, continued

For  $\langle p, q \rangle = \int_{-1}^1 pq$ , Example (Gram–Schmidt on  $1, x, x^2$ ) gave  $p_0 = 1$ ,  $p_1 = x$ ,  $p_2 = x^2 - \frac{1}{3}$ , with  $\langle p_0, p_0 \rangle = 2$ ,  $\langle p_1, p_1 \rangle = \frac{2}{3}$  and  $\langle p_2, p_2 \rangle = \frac{8}{45}$ . Use [Theorem 10.10.3](#) to obtain  $p_3$ , and check the recurrence reproduces  $p_2$ .

*Solution.* The weight  $w = 1$  is even on  $(-1, 1)$ , so every  $a_k = 0$  by the remark above. Then

$$b_1 = \frac{\langle p_1, p_1 \rangle}{\langle p_0, p_0 \rangle} = \frac{2/3}{2} = \frac{1}{3}, \quad b_2 = \frac{\langle p_2, p_2 \rangle}{\langle p_1, p_1 \rangle} = \frac{8/45}{2/3} = \frac{4}{15}.$$

The recurrence at  $k = 1$  gives  $p_2 = x \cdot x - \frac{1}{3} \cdot 1 = x^2 - \frac{1}{3}$ , which is what Gram–Schmidt produced. At  $k = 2$ ,

$$p_3 = x \left( x^2 - \frac{1}{3} \right) - \frac{4}{15} x = x^3 - \frac{3}{5} x.$$

*Check.*  $\langle p_3, 1 \rangle = \langle p_3, x^2 \rangle = 0$  because the integrands are odd, and

$$\langle p_3, x \rangle = \int_{-1}^1 \left( t^4 - \frac{3}{5} t^2 \right) dt = \frac{2}{5} - \frac{3}{5} \cdot \frac{2}{3} = 0.$$

So  $p_3 \perp \mathbb{R}[x]_{\leq 2}$ , as it should be. Two integrals produced a polynomial that Gram–Schmidt would have charged three for, and the saving grows with the degree.

## 10.10.4 Where the roots are

The roots of  $p_2 = x^2 - \frac{1}{3}$  are  $\pm 1/\sqrt{3} \approx \pm 0.577$ , and those of  $p_3 = x^3 - \frac{3}{5}x$  are 0 and  $\pm\sqrt{3/5} \approx \pm 0.775$ . All five lie inside  $(-1, 1)$ , where the weight lives. No coincidence. The argument first needs one fact about real polynomials, in which  $\text{mult}_r(f)$  is the multiplicity of  $r$  as a root of  $f$

(Definition 5.4.1).

**Lemma 10.10.4. Even multiplicities mean no sign change**

Let  $I \subseteq \mathbb{R}$  be an open interval and let  $f \in \mathbb{R}[x]$  be non-zero. If  $\text{mult}_r(f)$  is even for every  $r \in I$ , then  $f \geq 0$  throughout  $I$ , or  $f \leq 0$  throughout  $I$ .

*Proof.* By Theorem 5.4.3 over  $\mathbb{R}$ , write  $f = (x - d_1)^{m_1} \cdots (x - d_l)^{m_l} g$ , where  $d_1, \dots, d_l$  are the distinct real roots of  $f$ ,  $m_i = \text{mult}_{d_i}(f)$ , and  $g$  has no real root. Separate the factors according to whether  $d_i \in I$ :

$$f = h \cdot \left( \prod_{d_i \in I} (x - d_i)^{m_i} \right), \quad h = g \cdot \prod_{d_i \notin I} (x - d_i)^{m_i}.$$

Each  $m_i$  with  $d_i \in I$  is even by hypothesis, so the second factor is the square of a real polynomial and is  $\geq 0$  everywhere. The polynomial  $h$  has no root in  $I$ : its roots are the  $d_i$  outside  $I$ , and  $g$  has none. Being continuous and nowhere zero on the interval  $I$ ,  $h$  has constant sign there by the intermediate value theorem. Hence  $f$  has that same sign throughout  $I$ , weakly, with equality only at the  $d_i$  inside  $I$ . ■

**Theorem 10.10.5. Roots of an Orthogonal Polynomial**

Let  $w$  be a weight on  $(a, b)$ , let  $\langle p, q \rangle = \int_a^b p q w$ , and let  $(p_k)$  be the monic orthogonal polynomial sequence. For every  $k \geq 1$ , the polynomial  $p_k$  has  $k$  **distinct real** roots, and every one of them lies in the **open** interval  $(a, b)$ .

*Idea.* Suppose  $p_k$  changed sign at only a few points  $c_1, \dots, c_m$  of  $(a, b)$ , with  $m < k$ . Build the polynomial  $q = (x - c_1) \cdots (x - c_m)$ . It is cheap — degree at most  $k - 1$  — so  $p_k$  is orthogonal to it. But  $q$  was designed to change sign at exactly the points where  $p_k$  does, so the product  $p_k q$  changes sign nowhere, and a function of one sign that is not identically zero cannot integrate to zero against a positive weight. The contradiction forces  $m \geq k$ , and  $k$  distinct roots is all a polynomial of degree  $k$  can hold.

*Proof.* Fix  $k \geq 1$ . Let  $c_1 < \cdots < c_m$  be the roots of  $p_k$  in  $(a, b)$  whose multiplicity is **odd** (possibly  $m = 0$ ), and set  $q = (x - c_1) \cdots (x - c_m)$ , with  $q = 1$  when  $m = 0$ .

Suppose, for contradiction, that  $m \leq k - 1$ . Then  $q \in \mathbb{R}[x]_{\leq k-1}$ , so  $\langle p_k, q \rangle = 0$  by [Proposition 10.10.2](#).

Put  $f = p_k q$ , a non-zero polynomial (Theorem 0.8.3). Let  $r \in (a, b)$  be a root of  $f$ . By Theorem 5.4.2,  $\text{mult}_r(f) = \text{mult}_r(p_k) + \text{mult}_r(q)$ . If  $r = c_i$  for some  $i$ , then  $\text{mult}_r(q) = 1$  and  $\text{mult}_r(p_k)$  is odd by the choice of the  $c_i$ , so the sum is even. If  $r$  is none of the  $c_i$ , then  $q(r) \neq 0$ , so  $\text{mult}_r(q) = 0$  and  $\text{mult}_r(p_k)$  is even, again by the choice of the  $c_i$ . Either way  $\text{mult}_r(f)$  is even.

By [Lemma 10.10.4](#),  $f \geq 0$  throughout  $(a, b)$  or  $f \leq 0$  throughout  $(a, b)$ ; replacing  $q$  by  $-q$  if necessary, which changes neither  $\langle p_k, q \rangle = 0$  nor  $\deg q$ , we may assume  $f \geq 0$  there. Then  $fw$

is continuous and  $\geq 0$  on  $(a, b)$ , and it is  $> 0$  off the finitely many roots of  $f$ , because  $w > 0$ . By the calculus fact recalled when the inner product was set up — a continuous non-negative function with integral 0 vanishes identically — an integrand of that shape cannot integrate to 0. Therefore

$$\langle p_k, q \rangle = \int_a^b f(t) w(t) dt > 0,$$

contradicting  $\langle p_k, q \rangle = 0$ .

Hence  $m \geq k$ . Each  $c_i$  is a root of  $p_k$ , and  $p_k$  is non-zero of degree  $k$ , so  $m \leq k$  by Corollary 5.1.4; thus  $m = k$  and  $p_k$  has  $k$  distinct roots  $c_1, \dots, c_k$ , all in  $(a, b)$ . By Corollary 5.1.4 again,  $(x - c_1) \cdots (x - c_k)$  divides  $p_k$ , and both are monic of degree  $k$ , so  $p_k = (x - c_1) \cdots (x - c_k)$ . In particular each root is simple and  $p_k$  has no roots outside  $(a, b)$ , real or complex. This proves the theorem. ■

So the weight hands us  $k$  points inside its interval, chosen by the inner product rather than by us. The next subsection says what they are good for.

*Quick check.* The monic Laguerre polynomial of degree 2 for the weight  $w(t) = e^{-t}$  on  $(0, \infty)$  is  $p_2 = x^2 - 4x + 2$ . Check that [Theorem 10.10.5](#) is not lying.

*Solution.* The roots are  $2 \pm \sqrt{2}$ , that is  $\approx 0.586$  and  $\approx 3.414$ : real, distinct, and both in  $(0, \infty)$ . An unbounded interval is no obstacle, since the theorem asks only that the weight be positive with finite moments, and  $\int_0^\infty t^k e^{-t} dt = k!$ .

## 10.10.5 Gaussian quadrature

Chapter 5 wrote  $\int_0^1 p$  as a fixed combination of  $p(0)$ ,  $p(\frac{1}{2})$  and  $p(1)$ , valid for every  $p \in \mathbb{R}[x]_{\leq 2}$  (Example (Integrals and Slopes from Values)). The construction works for any prescribed nodes: integrate the Lagrange interpolation formula (Theorem 5.6.1) and  $k$  distinct nodes give a rule exact on  $\mathbb{R}[x]_{\leq k-1}$ .

But a  $k$ -point rule has  $2k$  numbers at its disposal,  $k$  nodes and  $k$  coefficients, and only  $k$  of them were spent. Spending the nodes too should buy degree  $2k - 1$ . Where to put them is the question, and the answer is the one place we have not chosen: the roots of  $p_k$ .

### Theorem 10.10.6. Gaussian Quadrature

Let  $w$  be a weight on  $(a, b)$  with  $\langle p, q \rangle = \int_a^b pqw$ , let  $k \geq 1$ , and let  $t_1 < \cdots < t_k$  be the roots of the monic orthogonal polynomial  $p_k$ , which lie in  $(a, b)$  by [Theorem 10.10.5](#). Let  $\ell_1, \dots, \ell_k \in \mathbb{R}[x]_{\leq k-1}$  be the Lagrange polynomials for these nodes (Definition 5.6.1) and set

$$\alpha_i = \int_a^b \ell_i(t) w(t) dt \quad (i = 1, \dots, k).$$

Then every  $\alpha_i$  is **strictly positive**, and

$$\int_a^b f(t) w(t) dt = \sum_{i=1}^k \alpha_i f(t_i)$$

for **every**  $f \in \mathbb{R}[x]_{\leq 2k-1}$ .

*Idea.* Two moves, one for each side of the identity. On the left, divide  $f$  by  $p_k$ :  $f = qp_k + r$ . The degree bookkeeping —  $\deg f \leq 2k - 1$  against  $\deg p_k = k$  — forces  $\deg q \leq k - 1$ , which is exactly the condition under which  $\int qp_k w = \langle p_k, q \rangle$  vanishes. So the integral of  $f$  is the integral of its remainder. On the right,  $p_k$  vanishes at every node, so the sum also only sees the remainder. And the remainder has degree at most  $k - 1$ , which is precisely the range on which  $k$  nodes reproduce a polynomial exactly. The two sides meet there.

*Proof.* Let  $f \in \mathbb{R}[x]_{\leq 2k-1}$ . Since  $p_k \neq 0$ , Theorem 5.1.1 gives  $q, r \in \mathbb{R}[x]$  with

$$f = qp_k + r, \quad \deg r < k.$$

If  $q = 0$  then  $q \in \mathbb{R}[x]_{\leq k-1}$  trivially. Otherwise  $qp_k = f - r$  has degree at most  $2k - 1$ , while  $\deg(qp_k) = \deg q + k$  by Theorem 0.8.3; hence  $\deg q \leq k - 1$ . Either way  $q \in \mathbb{R}[x]_{\leq k-1}$ , so Proposition 10.10.2 gives  $\langle p_k, q \rangle = 0$ , that is,  $\int_a^b qp_k w = 0$ . Therefore

$$\int_a^b fw = \int_a^b qp_k w + \int_a^b rw = \int_a^b rw. \quad (*)$$

At each node,  $p_k(t_i) = 0$ , so

$$f(t_i) = q(t_i)p_k(t_i) + r(t_i) = r(t_i) \quad (i = 1, \dots, k). \quad (**)$$

Since  $\deg r \leq k - 1$  and  $t_1, \dots, t_k$  are  $k$  distinct points, Theorem 5.6.1 (b) gives  $r = \sum_i r(t_i)\ell_i$ . Integrating against  $w$  and using the definition of  $\alpha_i$ ,

$$\int_a^b rw = \sum_{i=1}^k r(t_i) \alpha_i = \sum_{i=1}^k \alpha_i f(t_i),$$

the last step by (\*\*). Combining with (\*) proves the quadrature identity.

For positivity, fix  $j$  and apply the identity to  $f = \ell_j^2$ , which is legitimate because  $\deg \ell_j^2 = 2k - 2 \leq 2k - 1$ . By Theorem 5.6.1 (a),  $\ell_j(t_i) = \delta_{ij}$ , so the right-hand side collapses to  $\alpha_j$ , while the left-hand side is  $\langle \ell_j, \ell_j \rangle = \|\ell_j\|^2$ . Since  $\ell_j \neq 0$ , positive definiteness gives  $\alpha_j = \|\ell_j\|^2 > 0$ . ■

Each hypothesis did one job. Orthogonality to lower degrees killed the quotient on the left; the nodes being roots of  $p_k$  killed it again on the right. That is also why the nodes cannot be moved: shift one and the second cancellation fails. Exercise (C2: The ceiling is  $2k - 1$ ) shows  $2k - 1$  is a ceiling, not just what this proof happens to reach.

**Example. Two-point Gauss–Legendre**

Take  $w = 1$  on  $(-1, 1)$  and  $k = 2$ . Find the nodes and weights, and test the resulting rule on  $f = 1 + 2x + 3x^2 + 4x^3$  and on  $f = x^4$ .

*Solution.* The monic  $p_2$  is  $x^2 - \frac{1}{3}$  (Example (Gram–Schmidt on  $1, x, x^2$ )), so the nodes are  $t_1 = -1/\sqrt{3}$  and  $t_2 = 1/\sqrt{3}$ . The Lagrange polynomials are  $\ell_1 = (x - t_2)/(t_1 - t_2)$  and  $\ell_2 = (x - t_1)/(t_2 - t_1)$ , with  $t_1 - t_2 = -2/\sqrt{3}$ . Since  $\int_{-1}^1 t \, dt = 0$  and  $\int_{-1}^1 1 \, dt = 2$ ,

$$\alpha_1 = \frac{0 - 2t_2}{t_1 - t_2} = \frac{-2/\sqrt{3}}{-2/\sqrt{3}} = 1,$$

and  $\alpha_2 = 1$  by the same computation with the roles swapped. The rule is

$$\int_{-1}^1 f(t) \, dt = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right) \quad \text{for } f \in \mathbb{R}[x]_{\leq 3}.$$

Two evaluations, no coefficients to remember, every cubic exactly.

*The cubic test.* Exactly,  $\int_{-1}^1 (1 + 2t + 3t^2 + 4t^3) \, dt = 2 + 0 + 3 \cdot \frac{2}{3} + 0 = 4$ . By the rule,  $f(-1/\sqrt{3}) = 2 - \frac{10\sqrt{3}}{9}$  and  $f(1/\sqrt{3}) = 2 + \frac{10\sqrt{3}}{9}$ , whose sum is 4. The odd parts cancel and the constant and quadratic parts contribute  $1 + 1$  and  $1 + 1$ .

*The quartic test.* Exactly,  $\int_{-1}^1 t^4 \, dt = \frac{2}{5}$ . By the rule,  $(1/\sqrt{3})^4 + (-1/\sqrt{3})^4 = \frac{2}{9}$ . These differ, so degree  $4 = 2k$  is already out of reach. Compare Example (Integrals and Slopes from Values), where the nodes were fixed in advance and **three** of them guaranteed exactness only up to degree 2 — guaranteed by the construction, that is; that particular symmetric rule happens to integrate cubics exactly as well, which is luck of the symmetry and not something the construction promises.

*Warning.* **“Exact” means exact on polynomials.** Theorem 10.10.6 says nothing about  $\int_a^b f w$  for a general  $f$ , and nothing about what happens as  $k$  grows; on a non-polynomial the rule returns an approximation whose error is a question for analysis. What linear algebra delivers is exactness on a subspace, and the reason that subspace is the largest available.

## 10.10.6 The classical families

Four weights account for most of the orthogonal polynomials that occur in practice. The table records them; the entries are statements, and Exercise (B2: Two rows of the table) asks you to verify two of the rows by integration.

| Family    | Interval            | Weight $w(t)$      | $p_1$   | $p_2$               |
|-----------|---------------------|--------------------|---------|---------------------|
| Legendre  | $(-1, 1)$           | 1                  | $x$     | $x^2 - \frac{1}{3}$ |
| Chebyshev | $(-1, 1)$           | $(1 - t^2)^{-1/2}$ | $x$     | $x^2 - \frac{1}{2}$ |
| Hermite   | $(-\infty, \infty)$ | $e^{-t^2}$         | $x$     | $x^2 - \frac{1}{2}$ |
| Laguerre  | $(0, \infty)$       | $e^{-t}$           | $x - 1$ | $x^2 - 4x + 2$      |

The last two columns give the monic members of degrees 1 and 2;  $p_0 = 1$  in every row. The Chebyshev weight is unbounded at  $\pm 1$  and the Hermite and Laguerre intervals are unbounded, so in those three rows the integrals defining the inner product are improper; they converge, which is the finiteness condition our definition of a weight demanded. The first three weights are even on symmetric intervals, so their  $a_k$  all vanish and their members alternate in parity; the Laguerre weight is not, and  $a_0 = 1$  there. Section 9 of this chapter meets the Chebyshev family again from the Fourier side, where the substitution  $t = \cos \theta$  turns its awkward weight into  $d\theta$ .

## 10.10.7 Exercises

### 10.10.7.1 A. Check your understanding

#### Exercise. A1

- Define what it means for  $p_0, p_1, p_2, \dots$  to be a **monic orthogonal polynomial sequence** for an inner product on  $\mathbb{R}[x]$ .
- State the three-term recurrence, including the formulas for  $a_k$  and  $b_k$  and the starting values.
- True or false:  $b_k > 0$  for every  $k \geq 1$ . Justify your answer.
- For  $\langle p, q \rangle = \int_{-1}^1 pq$ , how many roots does  $p_7$  have, of what kind, and where?
- True or false: for **every** inner product on  $\mathbb{R}[x]$  one has  $\langle xp, q \rangle = \langle p, xq \rangle$ . Justify your answer.
- How many nodes does a Gaussian rule need in order to integrate every polynomial of degree at most 11 exactly?

#### Solution.

- A sequence with  $\deg p_k = k$  and leading coefficient 1 for every  $k$ , and  $\langle p_j, p_k \rangle = 0$  whenever  $j \neq k$  ([Definition 10.10.1](#)).
- With  $p_{-1} = 0$ ,  $p_0 = 1$  and  $p_1 = x - a_0$ , one has  $p_{k+1} = (x - a_k)p_k - b_k p_{k-1}$  for  $k \geq 1$ , where  $a_k = \langle xp_k, p_k \rangle / \langle p_k, p_k \rangle$  and  $b_k = \langle p_k, p_k \rangle / \langle p_{k-1}, p_{k-1} \rangle$  ([Theorem 10.10.3](#)).
- True. Both  $\langle p_k, p_k \rangle = \|p_k\|^2$  and  $\langle p_{k-1}, p_{k-1} \rangle = \|p_{k-1}\|^2$  are strictly positive, since  $p_k$  and  $p_{k-1}$  are non-zero and the inner product is positive definite.
- Seven roots, all real, all distinct, all in the open interval  $(-1, 1)$  ([Theorem 10.10.5](#)). In particular  $p_7$  has no complex roots and no repeated ones.
- False. Take the coefficient inner product  $\langle p, q \rangle = \sum_k u_k v_k$  on  $\mathbb{R}[x]$ , where  $p = \sum_k u_k x^k$  and  $q = \sum_k v_k x^k$  (Example (A functional with no Riesz vector)). Then  $\langle x \cdot 1, x \rangle = \langle x, x \rangle = 1$ , while  $\langle 1, x \cdot x \rangle = \langle 1, x^2 \rangle = 0$ . The identity is a property of the weight and node constructions ([Proposition 10.10.1](#)), not of inner products in general.
- Six. A  $k$ -point rule is exact up to degree  $2k - 1$ , and  $2 \cdot 6 - 1 = 11$ .

### 10.10.7.2 B. Practice

#### Exercise. B1: A weighted interval

On  $\mathbb{R}[x]$  put  $\langle p, q \rangle = \int_0^1 p(t)q(t)t \, dt$ . Compute the monic  $p_0$ ,  $p_1$  and  $p_2$ , and verify directly

that  $p_2 \perp \mathbb{R}[x]_{\leq 1}$ .

*Solution.* The moments are  $\langle x^k, 1 \rangle = \int_0^1 t^{k+1} dt = 1/(k+2)$ , so  $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$  for  $k = 0, 1, 2, 3$ .

$p_0 = 1$ . Then  $a_0 = \langle x, 1 \rangle / \langle 1, 1 \rangle = (1/3)/(1/2) = \frac{2}{3}$ , so  $p_1 = x - \frac{2}{3}$ , and

$$\langle p_1, p_1 \rangle = \frac{1}{4} - \frac{4}{3} \cdot \frac{1}{3} + \frac{4}{9} \cdot \frac{1}{2} = \frac{1}{4} - \frac{2}{9} = \frac{1}{36}.$$

Next  $\langle xp_1, p_1 \rangle = \int_0^1 (t^4 - \frac{4}{3}t^3 + \frac{4}{9}t^2) dt = \frac{1}{5} - \frac{1}{3} + \frac{4}{27} = \frac{2}{135}$ , so

$$a_1 = \frac{2/135}{1/36} = \frac{8}{15}, \quad b_1 = \frac{1/36}{1/2} = \frac{1}{18},$$

and [Theorem 10.10.3](#) gives

$$p_2 = \left(x - \frac{8}{15}\right)\left(x - \frac{2}{3}\right) - \frac{1}{18} = x^2 - \frac{6}{5}x + \frac{3}{10}.$$

*Verification.*  $\langle p_2, 1 \rangle = \int_0^1 (t^3 - \frac{6}{5}t^2 + \frac{3}{10}t) dt = \frac{1}{4} - \frac{2}{5} + \frac{3}{20} = 0$ , and  $\langle p_2, x \rangle = \int_0^1 (t^4 - \frac{6}{5}t^3 + \frac{3}{10}t^2) dt = \frac{1}{5} - \frac{3}{10} + \frac{1}{10} = 0$ . Since  $(1, x)$  spans  $\mathbb{R}[x]_{\leq 1}$ , this is the whole check.

### Exercise. B2: Two rows of the table

Verify the degree-1 and degree-2 entries of the table for two of the classical families.

- (a) Chebyshev:  $w(t) = (1 - t^2)^{-1/2}$  on  $(-1, 1)$ . You may use, without proof, that  $\int_{-1}^1 (1 - t^2)^{-1/2} dt = \pi$  and  $\int_{-1}^1 t^2 (1 - t^2)^{-1/2} dt = \pi/2$ .
- (b) Laguerre:  $w(t) = e^{-t}$  on  $(0, \infty)$ . You may use, without proof, that  $\int_0^\infty t^k e^{-t} dt = k!$ .

*Solution.*

- (a) The weight is even, so  $\langle x^j, x^i \rangle = 0$  whenever  $i + j$  is odd. Hence  $\langle x, 1 \rangle = 0$ , and  $x$  is monic of degree 1 and orthogonal to  $\mathbb{R}[x]_{\leq 0}$ , so  $p_1 = x$  by [Proposition 10.10.2](#). Write  $p_2 = x^2 + \beta x + \gamma$ . Then  $\langle p_2, x \rangle = \langle x^2, x \rangle + \beta \langle x, x \rangle + \gamma \langle 1, x \rangle = 0 + \beta \cdot \frac{\pi}{2} + 0$ , forcing  $\beta = 0$ ; and  $\langle p_2, 1 \rangle = \frac{\pi}{2} + 0 + \gamma\pi = 0$ , forcing  $\gamma = -\frac{1}{2}$ . So  $p_2 = x^2 - \frac{1}{2}$ .
- (b) The moments are  $\langle x^k, 1 \rangle = k!$ , so  $1, 1, 2, 6$  for  $k = 0, 1, 2, 3$ . For  $p_1 = x + \beta$ :  $\langle p_1, 1 \rangle = 1 + \beta = 0$ , so  $p_1 = x - 1$ . For  $p_2 = x^2 + \beta x + \gamma$ , the two conditions  $\langle p_2, 1 \rangle = 0$  and  $\langle p_2, x \rangle = 0$  read

$$2 + \beta + \gamma = 0, \quad 6 + 2\beta + \gamma = 0.$$

Subtracting gives  $\beta = -4$ , then  $\gamma = 2$ . So  $p_2 = x^2 - 4x + 2$ , whose roots  $2 \pm \sqrt{2}$  we already located in the Quick check above.

### Exercise. B3: One more Legendre polynomial

Continue [Example \(The monic Legendre polynomials, continued\)](#): compute  $\langle p_3, p_3 \rangle$  for  $p_3 = x^3 - \frac{3}{5}x$ , then use [Theorem 10.10.3](#) to find  $p_4$ . Verify that  $\langle p_4, 1 \rangle = \langle p_4, x^2 \rangle = 0$ .

*Solution.*

$$\langle p_3, p_3 \rangle = \int_{-1}^1 \left( t^6 - \frac{6}{5}t^4 + \frac{9}{25}t^2 \right) dt = \frac{2}{7} - \frac{6}{5} \cdot \frac{2}{5} + \frac{9}{25} \cdot \frac{2}{3} = \frac{2}{7} - \frac{6}{25} = \frac{8}{175}.$$

Since the weight is even,  $a_3 = 0$ , and  $b_3 = \langle p_3, p_3 \rangle / \langle p_2, p_2 \rangle = (8/175)/(8/45) = \frac{9}{35}$ . Hence

$$p_4 = x \left( x^3 - \frac{3}{5}x \right) - \frac{9}{35} \left( x^2 - \frac{1}{3} \right) = x^4 - \frac{6}{7}x^2 + \frac{3}{35}.$$

*Verification.*  $\langle p_4, 1 \rangle = \frac{2}{5} - \frac{6}{7} \cdot \frac{2}{3} + \frac{3}{35} \cdot 2 = \frac{14-20+6}{35} = 0$ , and

$$\langle p_4, x^2 \rangle = \frac{2}{7} - \frac{6}{7} \cdot \frac{2}{5} + \frac{3}{35} \cdot \frac{2}{3} = \frac{10-12+2}{35} = 0.$$

Orthogonality to  $x$  and  $x^3$  is automatic by parity.

#### Exercise. B4: A two-point rule on the unit interval

Work on  $(0, 1)$  with  $w = 1$ . Exercise B2 of Section 2 (Exercise (B2: Gram–Schmidt on polynomials)) found the monic orthogonal polynomials  $1, x - \frac{1}{2}, x^2 - x + \frac{1}{6}$ .

- (a) Write down the two-point Gaussian rule for  $\int_0^1 f$ .
- (b) Check it on  $f = x^3$ , and show that it fails on  $f = x^4$ .

*Solution.*

- (a) The nodes are the roots of  $x^2 - x + \frac{1}{6}$ , namely

$$t_{1,2} = \frac{1}{2} \mp \frac{\sqrt{3}}{6}, \quad t_2 - t_1 = \frac{\sqrt{3}}{3},$$

both in  $(0, 1)$ , as [Theorem 10.10.5](#) promises. With  $\ell_1 = (x - t_2)/(t_1 - t_2)$  and  $\int_0^1 t \, dt = \frac{1}{2}$ ,

$$\alpha_1 = \frac{\frac{1}{2} - t_2}{t_1 - t_2} = \frac{-\sqrt{3}/6}{-\sqrt{3}/3} = \frac{1}{2},$$

and  $\alpha_2 = \frac{1}{2}$  likewise. So  $\int_0^1 f = \frac{1}{2}(f(t_1) + f(t_2))$  for  $f \in \mathbb{R}[x]_{\leq 3}$ .

- (b) Write  $t_{1,2} = \frac{1}{2} \mp u$  with  $u = \sqrt{3}/6$ , so  $u^2 = \frac{1}{12}$ . The odd powers of  $u$  cancel in a sum over  $t_1$  and  $t_2$ , so

$$t_1^3 + t_2^3 = 2 \left( \frac{1}{8} + 3 \cdot \frac{1}{2} \cdot \frac{1}{12} \right) = 2 \cdot \frac{1}{4} = \frac{1}{2},$$

and half of that is  $\frac{1}{4} = \int_0^1 t^3 \, dt$ . For the quartic,

$$t_1^4 + t_2^4 = 2 \left( \frac{1}{16} + 6 \cdot \frac{1}{4} \cdot \frac{1}{12} + \frac{1}{144} \right) = 2 \cdot \frac{28}{144} = \frac{7}{18},$$

so the rule returns  $\frac{7}{36}$ , whereas  $\int_0^1 t^4 \, dt = \frac{1}{5}$ . The rule is exact to degree  $3 = 2k - 1$  and no further.

#### 10.10.7.3 C. Going deeper

**Exercise. C1: Interlacing roots**

Let  $w$  be a weight on  $(a, b)$  and let  $(p_k)$  be its monic orthogonal polynomial sequence. The roots of  $p_k$  are said to **interlace** those of  $p_{k+1}$  when exactly one root of  $p_k$  lies strictly between each pair of consecutive roots of  $p_{k+1}$ . This is true for every  $k$ ; prove the first two cases by hand.

- (a) Let  $r$  be the root of  $p_1$  and  $s_1 < s_2$  the roots of  $p_2$ . Prove that  $s_1 < r < s_2$ .
- (b) Let  $u_1 < u_2 < u_3$  be the roots of  $p_3$ . Prove that  $u_1 < s_1 < u_2 < s_2 < u_3$ .

*Hint: evaluate the three-term recurrence at a root.*

*Solution.*

- (a) By [Theorem 10.10.3](#) with  $k = 1$ ,  $p_2 = (x - a_1)p_1 - b_1p_0 = (x - a_1)(x - r) - b_1$ . Evaluating at  $r$  gives  $p_2(r) = -b_1 < 0$ , since  $b_1 > 0$ . By [Theorem 10.10.5](#),  $p_2 = (x - s_1)(x - s_2)$  with  $s_1 < s_2$  real, and this product is negative exactly on  $(s_1, s_2)$ . Hence  $s_1 < r < s_2$ .
- (b) By [Theorem 10.10.3](#) with  $k = 2$ ,  $p_3 = (x - a_2)p_2 - b_2p_1$ . At a root of  $p_2$  the first term dies, so

$$p_3(s_i) = -b_2 p_1(s_i) = -b_2(s_i - r) \quad (i = 1, 2).$$

By (a),  $s_1 - r < 0$  and  $s_2 - r > 0$ , and  $b_2 > 0$ , so  $p_3(s_1) > 0$  and  $p_3(s_2) < 0$ .

By [Theorem 10.10.5](#),  $p_3 = (x - u_1)(x - u_2)(x - u_3)$  with  $u_1 < u_2 < u_3$ . Such a product is negative on  $(-\infty, u_1)$ , positive on  $(u_1, u_2)$ , negative on  $(u_2, u_3)$ , and positive on  $(u_3, \infty)$ . So  $p_3(s_1) > 0$  puts  $s_1$  in  $(u_1, u_2)$  or in  $(u_3, \infty)$ , and  $p_3(s_2) < 0$  puts  $s_2$  in  $(-\infty, u_1)$  or in  $(u_2, u_3)$ . The option  $s_1 > u_3$  is impossible: it would give  $s_2 > s_1 > u_3$ , where  $p_3$  is positive, contradicting  $p_3(s_2) < 0$ . The option  $s_2 < u_1$  is impossible too: it would give  $s_1 < s_2 < u_1$ , where  $p_3$  is negative, contradicting  $p_3(s_1) > 0$ . Hence  $u_1 < s_1 < u_2$  and  $u_2 < s_2 < u_3$ , which is the assertion.

The general case is not proved this way. The coefficients  $a_k$  and  $b_k$  assemble into a symmetric tridiagonal matrix whose characteristic polynomial is  $p_k$ , and the interlacing then follows from the interlacing theorem for the eigenvalues of a symmetric matrix and its leading submatrices, which is Chapter 16's business.

**Exercise. C2: The ceiling is  $2k - 1$** 

Let  $w$  be a weight on  $(a, b)$  and let  $k \geq 1$ .

- (a) Let  $t_1, \dots, t_k$  be **any**  $k$  distinct real numbers and  $\alpha_1, \dots, \alpha_k$  **any** real numbers. Prove that there is an  $f \in \mathbb{R}[x]_{\leq 2k}$  with  $\int_a^b f w \neq \sum_i \alpha_i f(t_i)$ .
- (b) Hence deduce that the Gaussian rule of [Theorem 10.10.6](#) is exact on  $\mathbb{R}[x]_{\leq 2k-1}$  and on no larger space  $\mathbb{R}[x]_{\leq d}$ .

*Solution.*

- (a) Put  $g = (x - t_1) \cdots (x - t_k)$  and  $f = g^2$ , a polynomial of degree  $2k$ . Every node is a root of  $g$ , so  $f(t_i) = 0$  for every  $i$  and the right-hand side is 0, whatever the  $\alpha_i$  are. The left-hand side is  $\int_a^b g^2 w = \langle g, g \rangle$ , and  $g \neq 0$ , so  $\langle g, g \rangle > 0$  by positive definiteness.

The two sides differ.

- (b) Exactness on  $\mathbb{R}[x]_{\leq 2k-1}$  is [Theorem 10.10.6](#). Suppose the rule were exact on  $\mathbb{R}[x]_{\leq d}$  for some  $d \geq 2k$ . Since  $\mathbb{R}[x]_{\leq 2k} \subseteq \mathbb{R}[x]_{\leq d}$ , it would then be exact on every polynomial of degree at most  $2k$ . But its nodes are  $k$  distinct real numbers ([Theorem 10.10.5](#)), so (a) supplies a polynomial of degree  $2k$  on which it is not. Hence no such  $d$  exists, and  $2k - 1$  is the largest degree for which the rule is exact. The concrete instance  $k = 2$ ,  $f = x^4$  is the failure computed in [Example \(Two-point Gauss–Legendre\)](#).